CHAPTER 1

INTRODUCTION

1.1 Background

Partial Discharge is defined as a localized dielectric breakdown of a small portion of a solid or fluid electrical insulation system under high voltage stress, which does not completely bridge the electrodes (Lu et al., 2020). It typically originates from insulation defects such as voids, cracks, surface contamination, or imperfections within high-voltage electrical equipment (Agarwal et al., 2023; Hussain et al., 2021). These localized discharges degrade the insulation over time, potentially leading to insulation failure and catastrophic system breakdowns (Choudhary et al., 2024). Partial Discharge plays a pivotal role as an early warning indicator in various high-voltage components including transformers, cable terminations, switchgear, and Gas Insulated Switchgear (Govindarajan et al., 2023; Hussain et al., 2021). In particular, cross-linked polyethylene cables, widely adopted in countries like Malaysia for power distribution, are vulnerable to Partial Discharge-induced failures due to aging, improper jointing, or manufacturing defects (Choudhary et al., 2024; Rosle et al., 2021). The consequences of undetected Partial Discharge in such infrastructure can be severe, ranging from localized blackouts to significant financial losses and service disruption. Reliable detection and classification of Partial Discharge events are complicated by the transient and stochastic nature of the discharges, as well as the influence of environmental noise, electromagnetic interference, and the presence of multiple overlapping discharge sources (Baug et al., 2017; Ghosh et al., 2020). Traditional diagnostic techniques, such as visual inspection and periodic manual testing, are increasingly being replaced or augmented by automated systems capable of continuous monitoring and intelligent analysis (Sikorski et al., 2020).

The complexity of modern electrical networks and the increasing demand for uninterrupted power supply have driven the need for smarter, more resilient diagnostic frameworks (Vatsa et al., 2024). In this context, artificial intelligence, particularly machine learning and deep learning, have emerged as powerful tools for Partial Discharge signal interpretation (Long et al., 2024; Lu et al., 2020). These techniques offer capabilities for pattern recognition, noise filtering, and real-time classification, thereby enhancing the accuracy and speed of Partial Discharge diagnosis (Abdullah, Isa, et al., 2019; Song et al., 2018). As smart grids and condition-based monitoring systems become the norm, the development of robust, scalable, and noise-resilient Partial Discharge detection frameworks is imperative (Jia et al., 2020; J. Kim & Kim, 2021). These systems are not only critical to grid reliability and asset longevity but also align with global sustainability goals and digital grid modernization initiatives such as the EU Clean Energy Package and the U.S. Grid Modernization Initiative.

1.2 Significance of the Issue

Early and accurate detection and classification of Partial Discharge sources are paramount to ensuring the reliability and longevity of power systems (Kumar et al., 2024; Lu et al., 2020). A robust Partial Discharge monitoring system enables timely intervention and proactive maintenance, thereby preventing costly outages, equipment damage, and safety hazards. In safety-critical industries such as aviation and transportation, where electrical systems are vital, reliable Partial Discharge diagnosis becomes even more essential for operational efficiency and risk mitigation (Jiang et al., 2024; D. Wang et al., 2023).

1.2.1 Local Perspective

From an economic perspective, Partial Discharge detection and monitoring play a crucial role in reducing maintenance costs and preventing unexpected failures (Sarkar et al., 2021). The global Partial Discharge Monitoring Systems market was valued at \$529.91 million in 2023 and is projected to reach \$744.01 million by 2029, reflecting the growing awareness of Partial Discharge monitoring's importance in the electrical industry. The increasing demand for Partial Discharge monitoring systems in rapidly industrializing regions such as China and India highlights the necessity of reliable power infrastructure for economic growth (Song et al., 2018). Furthermore, government policies focusing on sustainable power generation and grid reliability emphasize the adoption of advanced monitoring techniques, including Partial Discharge detection, to enhance electrical infrastructure resilience (Abdullah, Isa, et al., 2019).

1.2.2 Market Size and Investment Trends in Partial Discharge Monitoring

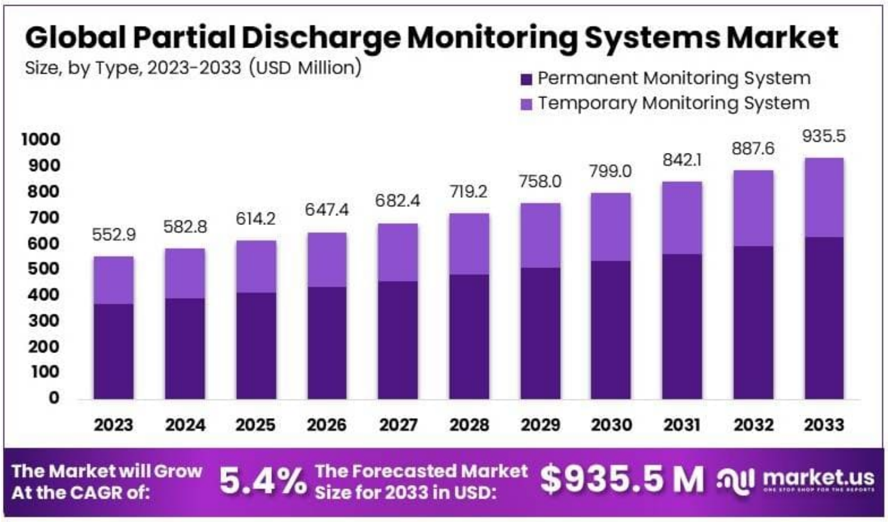

Figure 1.1. Global Partial Discharge Monitoring Systems Market by Type (2023–2033), showing growth from USD 552.9 million in 2023 to USD 935.5 million by 2033. (Market.us, 2024)

The global market for partial discharge monitoring systems is growing robustly, driven by the need for grid reliability and aging infrastructure. In 2023, the Partial Discharge monitoring market was valued around USD 550–580 million, and is projected to reach approximately USD 740–935 million over the next 5–10 years. This reflects a healthy compound annual growth rate on the order of 5–6%. Investments in smart grids and preventive maintenance are key factors propelling this growth (Jia et al., 2020; S. Kim et al., 2020). Notably, Asia Pacific is the leading region in Partial Discharge monitoring adoption, as countries like China, India, and Japan heavily invest in such systems to support their large power networks (Abdullah, Isa, et al., 2019; Song et al., 2018). North America and Europe also represent significant markets, with North America comprising roughly one-third of demand (about 35% in 2023) amid major grid modernization efforts. Malaysia, as part of the dynamic Asia Pacific region, is seeing increased attention to Partial Discharge monitoring through its national utility initiatives, although its domestic market size is a modest portion of the global figure (Chan et al., 2023). The trend worldwide is clear: utilities and industries are allocating more budget to Partial Discharge detection technology as part of broader condition-based maintenance and grid reliability programs (Sikorski et al., 2020).

From an investment standpoint, global utilities and governments are channeling substantial funds into grid monitoring technologies, including Partial Discharge systems. For example, the European Union has mandated regular Partial Discharge assessments and earmarked about €200 million specifically for upgrading infrastructure with advanced Partial Discharge monitoring by 2025. In the United States, utilities invested roughly USD 168 billion into the electric grid in 2023, with about USD 30.7 billion directed to transmission and USD 57.1 billion to distribution upgrades. A portion of these investments goes toward installing sensors and online monitoring (such as Partial Discharge detectors) to preempt failures (Liccardo et al., 2022). Government-backed programs like the U.S. Department of Energy's Grid Resilience and Innovation Partnerships, totaling $2.7 billion in funding, further illustrate the emphasis on modernizing the grid's monitoring and diagnostics capabilities. In Asia, state-owned utilities are following suit – State Grid China, for instance, has deployed innovative solutions like UAV-based ultrasonic Partial Discharge detection to patrol and maintain its vast transmission network (S. Kim et al., 2020). Overall, both private sector investments and public-sector funding mechanisms show a clear trend toward bolstering Partial Discharge monitoring as part of critical infrastructure development.

1.2.3 Economic Impact of Partial Discharge-Related Outages and Failures

Partial discharge is not just a technical issue; its consequences carry heavy financial costs. Partial Discharge activity deteriorates insulation over time and is a leading cause of electrical equipment failures, which in turn can trigger power outages (Agarwal et al., 2023; Hussain et al., 2021). Studies indicate that a large majority of disruptive power apparatus failures – on the order of 80–85% in high-voltage substations – are Partial Discharge-related (Sikorski et al., 2020). When such failures occur, the economic fallout can be significant. Outages mean downtime for industries, lost revenue for utilities, and recovery expenses, all of which add up quickly.

To put this in perspective, power outages cost businesses and economies billions annually. In the United States, for example, sudden power outages cost businesses over $150 billion each year. In Malaysia, a single major blackout (the 2003 nationwide outage) was estimated to have cost local businesses about USD 13.8 million. The table below highlights a few examples of Partial Discharge-related outage costs and their financial impact:

Even smaller-scale Partial Discharge incidents can incur tens of thousands of dollars in losses per day for the affected facility. A case study from a wind farm showed that partial discharge-induced tripping of a circuit breaker caused over $24,000 in lost revenue in just one occurrence. For utilities, a damaged transformer or switchgear that takes down a substation can mean costs reaching hundreds of thousands of dollars per day in unmet demand and repair efforts (Aslam et al., 2022). These direct costs are compounded by indirect losses: lost productivity, spoiled products, and cascading effects on other businesses and services. It's no surprise, then, that avoiding unplanned outages is a top financial priority for power providers and consumers alike.

Beyond individual incidents, chronic reliability issues have macroeconomic impacts. Frequent outages erode investor confidence and can stunt economic growth. South Asian economies, for instance, have been shown to suffer substantial GDP deficits due to power supply disruptions. In the corporate world, downtime can affect everything from manufacturing output to data center operations and safety systems. U.S. industry experts argue that many outage cost estimates actually understate the true impact, since they often fail to capture follow-on losses across supply chains and society. In short, Partial Discharge failures carry a hefty price tag, and this drives home the value of investing in Partial Discharge monitoring – the cost of prevention is far lower than the cost of unanticipated failure.

1.2.4 Government Funding and Policies on Partial Discharge Monitoring Infrastructure

Governments and regulators worldwide recognize the importance of grid reliability and have introduced policies and funding initiatives that, directly or indirectly, support Partial Discharge monitoring deployment. Malaysia's government, through the Energy Commission (Suruhanjaya Tenaga) and regulatory frameworks like Incentive-Based Regulation, has made reliability a key performance metric for utilities. Under the Incentive-Based Regulation framework, Tenaga Nasional Berhad – Malaysia's main utility – is incentivized to invest in infrastructure upgrades and preventive maintenance to minimize outages. This has led to continual investments in technologies such as online Partial Discharge detection, high-speed protection systems, and asset replacements to harden the grid (Abdullah, Isa, et al., 2019). The results are evident: Peninsular Malaysia's System Average Interruption Duration Index was just 48.13 minutes/customer in 2019, outperforming many developed countries. This world-class reliability is partly due to proactive identification of incipient faults (like Partial Discharge) before they cause breakdowns. The Malaysian government also supports research and development in grid monitoring through agencies like Tenaga Nasional Berhad Research and various university grants, fostering local expertise in Partial Discharge detection techniques (Chan et al., 2023).

Elsewhere in Asia, governments are bolstering grid resilience which includes Partial Discharge monitoring as a component. China's government, via State Grid and Southern Grid, has poured funding into modernizing its vast transmission network – including installing continuous monitoring sensors and even using drones for Partial

Discharge inspections (S. Kim et al., 2020). India has launched programs to upgrade its aging transmission and distribution infrastructure, backed by government financing and multilateral loans, which often stipulate the adoption of better monitoring to reduce technical losses and outages. Southeast Asian countries like Singapore and Thailand have also updated their grid codes and standards to encourage condition-based maintenance; utilities in these countries increasingly use Partial Discharge testing as part of equipment commissioning and routine checks (W. Wang et al., 2024). On the international stage, organizations such as the International Electrotechnical Commission have set standards (e.g. IEC 60270) that many governments adopt or reference, effectively making Partial Discharge measurement a normative requirement during factory testing and on-site commissioning of high-voltage equipment (Vera et al., 2023). Additionally, the World Bank and regional development banks fund electrification and grid reliability projects in developing nations, often including components for training and equipment to implement Partial Discharge monitoring as a means to extend asset life.

In advanced economies, regulatory pressure is perhaps most formalized in Europe and North America. Several European countries mandate periodic Partial Discharge testing for critical grid assets as part of their preventive maintenance regulations. The EU's focus on smart grids and its large funding programs (like the European Green Deal investments in energy networks) explicitly list grid diagnostic technologies eligible for support. In the United States, while no federal law specifically compels Partial Discharge monitoring, the trend is guided by reliability standards and incentives. The North American Electric Reliability Corporation has reliability standards that push utilities toward robust maintenance of key equipment (transformers, breakers, etc.), and Partial Discharge monitoring is recommended as a best practice in industry guidelines for preventing insulation failures (Hussain et al., 2021). U.S. federal funding through the Department of Energy's grid modernization initiatives and infrastructure bills has provided billions for grid upgrades, implicitly encouraging utilities to include advanced sensors (Partial Discharge, thermal, etc.) in their plans. In summary, government policies worldwide increasingly favor proactive maintenance and infrastructure resiliency, creating an environment where Partial Discharge monitoring is not just optional but often a supported or expected investment. This alignment of policy and funding with the goal of reducing blackouts has accelerated the adoption of Partial Discharge monitoring technology on a global scale.

1.2.5 Industry Initiatives and Partial Discharge Monitoring Programs

Electric utilities and industry players are at the forefront of implementing Partial Discharge monitoring as part of their asset management strategies. In Malaysia, both Tenaga Nasional Berhad and Sabah Electricity Sdn. Bhd. have launched initiatives to improve grid reliability that include partial discharge management. Tenaga Nasional Berhad, which serves Peninsular Malaysia, has been a regional leader in reliability and has systematically rolled out condition-based maintenance programs (Abdullah, Isa, et al., 2019). As part of these efforts, Tenaga Nasional Berhad pioneered Partial Discharge mapping techniques on its medium-voltage cables, conducting extensive field trials as early as the 2000s to evaluate their effectiveness. Today, Tenaga Nasional Berhad employs online partial discharge detection systems and Partial Discharge mapping in its operations – for example, to continuously monitor underground power cables and detect insulation defects before they escalate (Abdullah, Isa, et al., 2019). According to Tenaga Nasional Berhad's recent reports, integrating Partial Discharge monitoring has been instrumental in early fault identification and has helped mitigate the likelihood of major failures in the distribution network. Tenaga Nasional Berhad has extended these predictive maintenance practices to critical equipment like 33 kV switchgears, again using Partial Discharge sensing to preempt flashovers and catastrophic breakdowns (Alsumaidaee et al., 2022). By catching warning signs like Partial Discharge, Tenaga Nasional Berhad can schedule maintenance or component replacements proactively, thereby keeping its System Average Interruption Duration Index and System Average Interruption Frequency Index indices at impressively low levels. This approach directly translates to fewer outages and improved service continuity for customers.

Sabah Electricity Sdn. Bhd., which manages the grid in Sabah, has also recognized the importance of such monitoring, albeit starting from a more challenging reliability baseline. Sabah historically faced frequent power disruptions – in 2005, System Average Interruption Duration Index in Sabah was over 4,000 minutes per customer (reflecting very poor reliability). Through various programs, Sabah Electricity Sdn. Bhd. has dramatically improved this, reducing System Average Interruption Duration Index to about 274 minutes in 2023 and setting a target of 100 minutes by 2030. To achieve this, Sabah Electricity Sdn. Bhd. is upgrading its infrastructure and maintenance regimes. While specific Partial Discharge monitoring programs at Sabah Electricity Sdn. Bhd. are less publicized than Tenaga Nasional Berhad's, the utility is known to be collaborating with Tenaga Nasional Berhad and technical partners to adopt similar best practices. This likely includes using portable Partial Discharge detection units for surveying substations and cables and prioritizing the replacement of equipment that shows Partial Discharge symptoms (Sikorski & Gielniak, 2025). The Malaysian government has been supportive, with federal funds aiding Sabah's grid improvements (including a subsidy to cover Sabah Electricity Sdn. Bhd.'s operational deficits while upgrades are underway). Industry-wide, there is also knowledge sharing: Tenaga Nasional Berhad Research and local universities have worked on Partial Discharge detection techniques suited for the tropical environment (where high humidity can exacerbate surface discharge) (Azam et al., 2023). These R&D outputs benefit Sabah Electricity Sdn. Bhd. and other regional utilities in crafting their Partial Discharge monitoring and mitigation strategies.

Beyond the utilities, industry involvement includes equipment manufacturers and service providers who run Partial Discharge monitoring programs or services. Companies like Doble, Megger, Qualitrol, and EA Technology offer Partial Discharge testing as part of substation maintenance contracts and have conducted training workshops in Malaysia and across Asia to build local capacity. Some large industrial consumers (e.g. semiconductor factories or oil & gas facilities in Malaysia) have also started installing Partial Discharge monitors on their critical high-voltage equipment to avoid costly downtime. For instance, continuous Partial Discharge monitoring systems are now available

Figure 1.2. Example of partial discharge detection using an acoustic imaging camera (Fluke ii910), where colored contours indicate Partial Discharge intensity on substation bushings. (Fluke, n.d.)

for large motors, generators, and gas-insulated switchgear – and vendors often cite success stories where their systems gave early warnings that averted a failure (Y.-T. Chen et al., 2018; Seri et al., 2021). Global industry statistics underscore the value of Partial Discharge monitoring: one supplier notes that 85% of disruptive substation failures are Partial Discharge-related, meaning a Partial Discharge monitoring program can significantly reduce the risk of those failures (Sikorski et al., 2020). Another industry report highlighted that insulation failure (often traceable to Partial Discharge) contributes to up to 90% of switchgear defects, reinforcing why so many maintenance teams now include Partial Discharge checks as a standard procedure (Alsumaidaee et al., 2022).

In the Malaysian context, the collaboration between industry and the national utility is evident in programs like the Intelligent Predictive & Diagnostic Monitoring system mentioned by Tenaga Nasional Berhad, which integrates Partial Discharge sensors in the broader smart grid monitoring platform. Tenaga Nasional Berhad and Sabah Electricity Sdn. Bhd. also participate in regional forums (through ASEAN utilities and international bodies) to share incident statistics and lessons. While detailed Partial Discharge incident statistics in Malaysia are not publicly released, anecdotal evidence and case reports suggest that numerous minor faults have been caught and resolved thanks to Partial Discharge detection, preventing them from becoming major outages. Each avoided incident contributes to improved reliability metrics. The drop in Tenaga Nasional Berhad's transmission system minutes (only 0.27 minutes of interruption in 2019 for Peninsular Malaysia) is partly credited to such preventive monitoring. Regionally, countries like Singapore report very low outage rates as well, attributable to stringent monitoring and maintenance regimens (which include Partial Discharge testing). This industry-wide push – from utility engineers patrolling with ultrasonic Partial Discharge detectors, to control centers analyzing continuous online Partial Discharge data – highlights a robust commitment to nipping problems in the bud (Abdullah, Isa, et al., 2019; Alshalawi & Al-Ismail, 2024).

1.3 Existing Approaches and Challenges

The diagnosis of Partial Discharge (PD) is a critical aspect of condition monitoring for high-voltage equipment, providing essential insights into the health of insulation systems (Hussain et al., 2021). Diagnostic approaches have evolved from traditional methods, which rely on the manual interpretation of discharge patterns, to modern automated systems employing artificial intelligence and machine learning (Mas'ud et al., 2016). While conventional techniques have been foundational, they are often limited by subjectivity, inefficiency in handling large datasets, and susceptibility to noise, which can compromise diagnostic accuracy (Das et al., 2023; Florkowski, 2021).

To address this issue, automated methods have been developed to provide more objective and efficient analysis (Lu et al., 2020). However, these advanced systems introduce their own set of challenges, including the complexity of robust feature extraction, the need for effective noise cancellation, and ensuring model generalization across varied operational conditions (Friebe & Jenau, 2024; Long et al., 2024).

1.3.1 Limitations of Traditional Methods

Conventional methods for diagnosing Partial Discharge phenomena primarily rely on the manual interpretation of Phase-Resolved Partial Discharge patterns by skilled operators (Florkowski, 2021; Kasinathan et al., 2017). This approach, while historically effective, is increasingly recognized for its inherent limitations, which hinder its applicability in modern high-voltage insulation systems.

One of the foremost challenges in manual Partial Discharge analysis is its subjectivity. The interpretation of Phase-Resolved Partial Discharge patterns depends heavily on the experience and expertise of the operator, making it susceptible to human error (Lu et al., 2020; Mas'ud et al., 2016). Variations in individual judgment can lead to inconsistencies in diagnosis, ultimately affecting the reliability of the results (Kunicki et al., 2018). Such subjectivity poses a significant risk in critical applications where precise Partial Discharge identification is essential for ensuring equipment longevity and operational safety.

Another major drawback is the time-consuming nature of manual Partial Discharge analysis. Evaluating large volumes of Phase-Resolved Partial Discharge data requires substantial effort and expertise, making the process inefficient, particularly for large-scale industrial applications (Long et al., 2024). As the demand for faster and more efficient Partial Discharge diagnosis grows, the reliance on human interpretation becomes increasingly impractical, emphasizing the need for automated techniques that can process large datasets more efficiently (Das et al., 2023).

Furthermore, traditional Partial Discharge diagnosis methods are highly susceptible to noise interference, which poses a significant challenge in real-world applications (Friebe & Jenau, 2024). In practical scenarios, especially in on-site measurements, Partial Discharge signals are often masked by background noise, making it difficult to distinguish genuine Partial Discharge activity from external disturbances (Bao et al., 2024; Ghosh et al., 2020). Conventional filtering techniques are not always effective in isolating Partial Discharge signals, leading to potential misclassification or overlooked defects (Hussein et al., 2016). The presence of noise-induced errors further complicates the reliability of manual Phase-Resolved Partial Discharge pattern interpretation.

In addition to noise susceptibility, conventional techniques struggle to handle complex Partial Discharge scenarios where multiple Partial Discharge sources coexist within an insulation system (Baug et al., 2017). In such cases, overlapping Partial Discharge patterns make it challenging to extract meaningful features and perform accurate classification (Biswas et al., 2016; Firuzi et al., 2017). Traditional methods often lack the capability to effectively differentiate between multiple Partial Discharge sources, limiting their diagnostic accuracy and effectiveness. This limitation is particularly problematic in aging power equipment, where different types of insulation defects may simultaneously contribute to Partial Discharge activity.

1.3.2 Challenges with Automated Methods

Given these challenges discussed in Section 1.3.1, there is a growing need for advanced Partial Discharge diagnosis techniques that can overcome the limitations of manual interpretation. The integration of machine learning and signal processing methods has shown promising potential in addressing these issues by providing objective, efficient, and noise-resilient Partial Discharge classification (Lu et al., 2020; Mas'ud et al., 2016). These approaches offer significant advantages, including improved efficiency, objectivity, and the ability to process large volumes of data. However, despite their potential, automated methods still face several fundamental challenges that must be addressed to achieve reliable and accurate Partial Discharge classification.

One of the primary challenges in automated Partial Discharge classification is the process of feature extraction. Effective Partial Discharge analysis requires the selection of meaningful features that can accurately represent different types of discharge activity (Long et al., 2024). However, Partial Discharge signals are complex and often exhibit highly stochastic behavior, making it difficult to determine which features are most relevant for classification (Dai et al., 2017; X. Li et al., 2018). The process of selecting and engineering features typically requires significant domain expertise to ensure that extracted parameters adequately capture the characteristics of different Partial Discharge types (W. Zhou et al., 2017). Without careful feature selection, machine learning models may struggle to distinguish between genuine Partial Discharge signals and external noise, leading to misclassification and unreliable diagnostic outcomes.

Another significant obstacle is the optimization of machine learning algorithms for Partial Discharge classification. Selecting the most suitable algorithm is not always straightforward, as different techniques exhibit varying levels of effectiveness depending on the nature of the Partial Discharge data (Fei et al., 2024). The stochastic nature of Partial Discharge signals further complicates this process, as the variability in signal characteristics makes it difficult to establish a universally optimal model (Vigneshwaran et al., 2022). Furthermore, machine learning algorithms often require fine-tuning of hyperparameters to achieve high classification accuracy. This optimization process can be computationally intensive and may necessitate extensive trial-and-error experimentation to identify the best-performing configurations.

Beyond algorithm selection and optimization, a major concern with automated Partial Discharge classification is the issue of generalization. Many machine learning models struggle to generalize well when applied to new or unseen data (Raymond et al., 2022). This problem is particularly pronounced when models are trained on limited or imbalanced datasets, where certain Partial Discharge types may be underrepresented (Tuyet-Doan et al., 2021). As a result, the trained models may exhibit high accuracy on the training data but fail to maintain their performance when applied to real-world scenarios. The challenge of generalization necessitates the use of robust data augmentation techniques, transfer learning, or adaptive learning mechanisms to improve model resilience across different operating conditions.

Additionally, the presence of environmental noise and interference remains a significant challenge for automated Partial Discharge classification systems (Azam et al., 2023). In practical applications, Partial Discharge signals are often contaminated by external noise sources, making it difficult to isolate and identify discharge patterns accurately (Friebe & Jenau, 2024). The effectiveness of an automated classification system heavily depends on the robustness of its feature extraction and noise filtering techniques. Advanced signal processing methods, such as wavelet transforms, adaptive filtering, and deep learning-based denoising techniques, have been explored to mitigate the impact of noise (Abdullah, Isa, et al., 2019; T. Jin et al., 2019). However, ensuring the reliability of Partial Discharge classification in noisy environments remains an active area of research.

1.4 Problem Formulation

1.4.1 Research Gap

Current research in partial discharge detection for overhead power lines often focuses on specific datasets or individual algorithms, lacking a comprehensive comparison of traditional and modern machine learning techniques across diverse datasets (Basharan et al., 2018; Song et al., 2018). This limitation restricts the understanding of the relative strengths and weaknesses of different algorithms under varying conditions, such as noise levels, environmental factors, and signal acquisition setups. Performance reported on a single dataset may not generalize well to other datasets due to differences in partial discharge signal characteristics, noise profiles, and measurement protocols, necessitating a broader comparison across multiple datasets to assess the robustness and generalizability of different machine learning techniques. The optimal selection and combination of features for partial discharge detection also remain underexplored, with existing studies rarely investigating the impact of different feature sets, such as time-domain, frequency-domain, and deep-learned embeddings, on the performance of various machine learning models (Long et al., 2024). Different algorithms may be sensitive to specific feature types, and optimizing feature selection can significantly enhance detection accuracy and efficiency, making a systematic investigation of feature combinations crucial for identifying the most informative and discriminative features for each algorithm. This gap hinders the development of robust and generalizable partial discharge detection systems that can perform reliably across diverse overhead power line configurations and environmental conditions, and the lack of comprehensive algorithm comparisons and feature analyses makes it challenging to select the most appropriate algorithm and feature set for specific applications. This study aims to address this gap by providing a comparative analysis of traditional and modern machine learning methods across multiple datasets and investigating the impact of different feature combinations on their performance.

1.4.2 Problem Statement

Based on the identified research gap, this study addresses several key problems. First, there is a lack of a comprehensive comparison of machine learning algorithms for partial discharge detection, which includes evaluating the performance of traditional algorithms (Support Vector Machine, Random Forest, K-Nearest Neighbors, Naive Bayes, AdaBoost) in terms of accuracy, precision, recall, F1-score, Area Under the Curve, and computational cost, as well as understanding the trade-offs between performance and computational complexity, which is critical for real-time partial discharge monitoring in overhead power lines (Kumar et al., 2024; Lu et al., 2020). Second, there is limited understanding of the optimal feature sets for different machine learning algorithms in partial discharge detection, requiring an investigation of various feature extraction techniques, including time-domain features (e.g., pulse amplitude, width), frequency-domain features (e.g., Fast Fourier Transform, wavelet transforms), and deep-learned embeddings, to evaluate their impact on the performance of each algorithm and identify the most discriminative features that contribute significantly to accurate partial discharge classification (Long et al., 2024). Third, there is a need to evaluate the generalizability of partial discharge detection models across diverse datasets, which involves testing trained models on datasets not used during training, representing different noise conditions, environmental factors, and signal acquisition protocols, to provide insights into the robustness and real-world applicability of the developed models for overhead power line monitoring (Raymond et al., 2022).

1.5 Study Objectives

Based on the hypotheses, this study aims to achieve the following objectives:

1. To compare the performance of various traditional machine learning algorithms (Support Vector Machine, Decision Tree, Random Forest, K-Nearest Neighbors, ANN) for partial discharge detection.
2. To identify the optimal feature set for each machine learning algorithm by evaluating different feature combinations.
3. To assess the generalizability of the trained models by testing their performance on a combined dataset composed of multiple individual Partial Discharge datasets.

1.6 Scope Of Work

The scope of this thesis includes:

1. Preprocessing and feature extraction from publicly available partial discharge datasets. This involves applying signal conditioning techniques to remove noise and irrelevant components, followed by the extraction of features such as wavelet coefficients, phase-resolved partial discharge patterns, statistical descriptors, and entropy measures. Feature selection will be guided by existing literature and optimized to capture the characteristics of Partial Discharge activity under different conditions.
2. Training and evaluation of traditional machine learning models using cross-validation on individual datasets and a combined dataset. This includes training each selected machine learning algorithm on individual Partial Discharge datasets using k-fold cross-validation to ensure statistical robustness. The trained models will then be evaluated on separate test sets as well as on the aggregated dataset to measure generalization performance.
3. Comparison of model performance based on accuracy, precision, recall, F1-score, and computational time. These performance metrics will allow for a comprehensive evaluation of each algorithm, accounting for both predictive quality and com-

putational efficiency, which are critical factors in practical deployment.

4. Analysis of the impact of different feature combinations on the performance of each model. This involves systematic evaluation of various feature subsets and quantifying their influence on the classification results. The goal is to identify optimal feature sets that maximize detection accuracy while minimizing computational costs, thus supporting real-time monitoring applications.

1.7 Organisation Of Thesis

This thesis is organized as follows: Chapter 1 introduces the background, motivation, problem statement, hypotheses, objectives, and scope of the research. It provides a comprehensive overview of the context, rationale, and goals of the study, setting the stage for the subsequent chapters.

Chapter 2 reviews the existing literature on partial discharge detection, covering various methods and challenges. This chapter provides a detailed analysis of prior research in partial discharge analysis, including traditional machine learning techniques, signal processing approaches, and recent developments in automated diagnostics. It identifies current gaps and motivates the research direction of this study.

Chapter 3 details the methodology employed, including dataset acquisition, signal preprocessing, feature extraction, model selection, and performance evaluation. This chapter outlines the entire experimental pipeline, ensuring clarity and reproducibility. It describes the datasets used, the preprocessing strategies applied, the features extracted from partial discharge signals, and the machine learning algorithms selected for evaluation.

Chapter 4 presents the experimental results, comparing the performance of various machine learning algorithms and analyzing the effects of feature selection. This chapter includes comparative results across multiple datasets, evaluates generalization, and interprets model behavior in relation to feature sets.

Chapter 5 concludes the study by summarizing the key findings, discussing their implications for real-world partial discharge monitoring, and suggesting directions for future research. It highlights the contributions of the work and outlines opportunities to extend the study, such as integration with Internet of Things sensors or hybrid deep learning models.

The Appendix includes supplementary materials such as dataset details, signal transformation formulas, algorithmic configurations, and extended experimental results. This section serves as a technical reference for readers and researchers interested in replicating or building upon this work.

CHAPTER 2

LITERATURE REVIEW: PARTIAL DISCHARGE DETECTION USING TRADITIONAL MACHINE LEARNING

2.1 Machine Learning in Partial Discharge Detection

The application of machine learning (ML) techniques to partial discharge (PD) analysis has become increasingly prevalent, driven by the need to overcome the limitations inherent in conventional diagnostic methods (Lu et al., 2020; Mas'ud et al., 2016). Historically, PD assessment relied heavily on the manual interpretation of measurement data, such as Phase-Resolved Partial Discharge (PRPD) patterns, by human experts. This process is often subjective, time-consuming, and can be unreliable, particularly when dealing with the large volumes of data generated by modern online monitoring systems (Alsumaidaee et al., 2022; Seri et al., 2021). Traditional approaches also face significant challenges in noisy operational environments, where distinguishing PD signals from external interference and background noise is a primary obstacle (Ghosh et al., 2020; Park et al., 2023). Moreover, identifying and separating signals from multiple concurrent PD sources (Baug et al., 2017; Firuzi et al., 2017) and reliably quantifying the severity or danger of an insulation defect remain complex tasks for conventional analysis (Ma et al., 2021; Orellana et al., 2023).

Machine learning offers a pathway to automated, objective, and more accurate PD analysis, forming the core of advanced condition monitoring and diagnostic systems (Kumar et al., 2024; Lu et al., 2020). A primary application of ML in this field is the classification of PD source type such as differentiating between corona, surface discharges, and internal voids which is essential for assessing the risk posed to insulation integrity (Anjum et al., 2017; Borghei & Ghassemi, 2022). Another key task is anomaly detection, where ML models are trained to identify PD events as deviations from normal operational signals or ambient noise, a capability crucial for robust online monitoring (Florkowski, 2021; Thi et al., 2020). Furthermore, ML algorithms are increasingly used for the localization of PD sources, estimating their physical position within high-voltage (HV) equipment, which facilitates targeted maintenance and repair (Chan et al., 2023; Liu et al., 2020). The overarching motivation for employing ML is to develop intelligent systems that can learn complex patterns directly from sensor data, adapt to changing operational conditions, and provide consistent and reliable diagnostic assessments (Sikorski & Gielniak, 2025).

2.2 Overview of Machine Learning Algorithms for Partial Discharge Detection

A variety of traditional machine learning algorithms have been employed for Partial Discharge (PD) detection and analysis. These algorithms are primarily used for tasks such as the classification of PD types, localization of PD sources, and noise rejection. This section provides an overview of some commonly used traditional ML algorithms found in the reviewed literature.

2.2.1 Support Vector Machines (SVM)

Support Vector Machines (SVMs) are supervised learning models that analyze data for classification and regression. In classification tasks, SVMs construct an optimal hyperplane in a high-dimensional space to serve as a decision boundary with the largest possible margin to the nearest data points of any class.

SVMs have been widely used for classifying PD sources and defect types. For instance, SVM has been effectively utilized to recognize PDs in switchgear from ultra-high frequency (UHF) signal data (H. Jin et al., 2018) and to classify four different PD types using an optimized model (Dai et al., 2017). The versatility of SVMs is further demonstrated in their application to fault diagnosis in Gas-Insulated Switchgear (GIS) using

Zernike moment features (Z. Li et al., 2022) and for classifying PD signals captured with a Hilbert fractal antenna (Zahed et al., 2017). They are also used for localizing PD sources in GIS based on image edge detection features (X. Li et al., 2019). Advanced applications include using a Brain Storm Optimization (BSO) enhanced SVM for pattern recognition of ultrasonic PD signals in transformers (Y. Zhou et al., 2022).

The selection of SVM is often motivated by its strong theoretical foundation, effectiveness in high-dimensional spaces, and robust performance even with limited training data (Bin et al., 2020; W. Zhou et al., 2017). Its ability to handle the complex characteristics of PD data and its strong performance in multi-class classification make it a preferred choice for pattern recognition and distinguishing between different PD types (H. Jin et al., 2018; Serttaş & Hocaoğlu, 2020). Furthermore, One-Class SVM (OCSVM) has been chosen for its ability to model normal, PD-free conditions without requiring a fully labeled training set, which is advantageous for online monitoring systems operating in low signal-to-noise ratio (SNR) environments (Parrado-Hernández et al., 2018).

In terms of performance, an improved SVM achieved a PD diagnosis accuracy of 91.23% (Z. Li et al., 2022), while another study reported a 97% recognition rate for classifying PD types (Zahed et al., 2017). For defect classification in ceramic insulators, SVM outperformed k-NN, Naive Bayes, and Decision Tree with 90.6% accuracy (Ma et al., 2021). For aging state recognition in polyethylene-insulated cables, SVM achieved up to 100% accuracy with a small subset of features (Morette et al., 2020).

2.2.2 Random Forest (RF)

Random Forest (RF) is an ensemble learning method that constructs a multitude of decision trees during training. It outputs the class selected by the most trees for classification tasks, making it highly regarded for its robustness against overfitting and its ability to manage high-dimensional data.

RF has been successfully applied in various PD detection scenarios, including identifying the location of PD sources (Bag et al., 2019) and classifying PD types (Guo et al., 2021; Prabhu et al., 2016). The algorithm has also been used to identify faults based on decomposition gases (Zhu et al., 2025). Specific applications include locating PD sources using features from optical sensor data (Bag et al., 2019), identifying faults based on the concentration of decomposition gases (Tan et al., 2022), and classifying defects in on-board cable terminals for high-speed railways (Guo et al., 2021).

The choice of RF is often driven by its high accuracy, its ability to process large datasets with numerous input variables, and its built-in feature ranking capabilities (Bag et al., 2019). Researchers have also developed enhanced versions, such as a hypergraph-based improved RF algorithm (HG-RF-RFE) for better feature elimination and classification (Govindarajan et al., 2021) and a Dung Beetle Algorithm optimized Random Forest (DBO-RF) to improve defect detection in GIS (Zhu et al., 2025).

In terms of performance, RF has consistently demonstrated high accuracy. A PD source location accuracy of 95.6% was achieved using RF on optical sensor data (Bag et al., 2019). The HG-RF-RFE technique yielded an impressive 98.8% accuracy for PD source classification in both noise-free and noisy conditions (Govindarajan et al., 2021), while the DBO-RF model reached 92.2% accuracy in detecting GIS defects (Zhu et al., 2025).

2.2.3 Decision Trees (DT)

Decision Trees are a non-parametric supervised learning method used for classification and regression. They build a model that predicts a target variable's value by learning simple, interpretable decision rules inferred from data features. Decision trees have been applied for PD fault type recognition (Lee & Horng, 2020; Zeng et al., 2019) and for detecting insulation faults in transformers, where they have shown high accuracy for specific fault classes (Maseko et al., 2025). These models are often chosen for their interpretability and their capacity to handle both numerical and categorical data. For instance, a decision tree evaluation model was constructed to provide a clear framework for diagnosing PD conditions in DC gas-insulated equipment (GIE) (Zeng et al., 2019).

They have also served as a baseline model for comparison against more complex algorithms like Back Propagation Neural Networks (BPNN) (Fei et al., 2024). In terms of performance, a Fuzzy Logic Clustering Decision Tree (FLCDT) was shown to perform significantly better than standard decision tree algorithms like CART and See5 in terms of classification precision for transformer diagnostics (Lee & Horng, 2020). The decision tree model used for PD diagnosis based on decomposition gases was noted for offering a structured and effective diagnostic framework (Zeng et al., 2019). Furthermore, an accuracy of 100% was achieved for specific insulation faults, such as those caused by thermal aging, using Decision Trees (Maseko et al., 2025).

2.2.4 k-Nearest Neighbors (k-NN)

k-Nearest Neighbors (k-NN) is a simple, instance-based learning algorithm. For classification, an object is assigned to the class most common among its k k nearest neighbors, as determined by a majority vote. k-NN has been applied to classify PD types in GIS (Bin et al., 2020; Fei et al., 2024) and for online stator winding fault diagnosis in induction motors (Sarkar et al., 2021). The algorithm is frequently chosen for its simplicity, effectiveness, and ease of implementation, especially in scenarios with irregular decision boundaries (Bin et al., 2020). It has also been noted for enabling effective online fault detection with high prediction accuracy for motor faults (Sarkar et al., 2021).

In terms of performance, k-NN achieved an accuracy of 88.5% in classifying four types of PD in GIS (Bin et al., 2020). In motor fault diagnosis, it demonstrated acceptable prediction accuracy, making it viable for online monitoring systems (Sarkar et al., 2021). While serving as a baseline model, k-NN's results have been compared against ANNs that achieved classification accuracies as high as 96.03% (Haiba & Eliwa Gad, 2024).

2.2.5 Artificial Neural Networks (ANN) (Traditional Architectures)

Artificial Neural Networks (ANNs), in their traditional forms like multilayer perceptrons (MLPs) trained with back-propagation, are computational models inspired by the brain's structure. These models consist of interconnected nodes arranged in layers. Other traditional ANN architectures used in PD detection include Self-Organizing Maps (SOM), which are used for clustering and visualization, and Learning Vector Quantization (LVQ) networks.

ANNs have been widely applied for PD pattern recognition and classification of defect types. For example, ANNs have been used to classify discharges from different defects in ceramic insulators (Anjum et al., 2017) and to diagnose faults in motor stator windings (Y.-T. Chen et al., 2018). A Back-Propagation ANN (BP-ANN) has been used for PD identification in GIS based on acoustic emission signals (Hou et al., 2021), and a Learning Vector Quantization (LVQ) network has been used for localizing PD in noisy environments by analyzing winding sections (Mohammadirad et al., 2025).

ANNs are selected for their ability to model complex, non-linear relationships between input features and output classes, as well as their capacity for learning directly from data (Y.-T. Chen et al., 2018; Hou et al., 2021). They have demonstrated high recognition rates for various PD faults and the capability to determine degradation stages (Mas'ud et al., 2016). In terms of performance, recognition rates exceeding 95% were reported for each defect class using ANNs (Anjum et al., 2017). An accuracy of 90% was achieved in diagnosing stator faults (Y.-T. Chen et al., 2018), and 96.03% accuracy was obtained for insulator defect classification using discrete wavelet transform features (Haiba & Eliwa Gad, 2024). A study classifying PD types reported a 97% recognition rate (Zahed et al., 2017), and a BP-ANN model for GIS defects achieved 93.7% accuracy (Hou et al., 2021).

2.2.6 Fuzzy Logic Based Classifiers

Fuzzy logic systems operate on ”degrees of truth” rather than binary true/false logic, making them well-suited for managing the uncertainty and imprecision inherent in real-world data. Common fuzzy logic-based classifiers include Fuzzy Decision Trees, Fuzzy C-Means (FCM) clustering, and Adaptive Neuro-Fuzzy Inference Systems (ANFIS). Fuzzy logic has been applied to diagnose PD in cast-resin transformers (Lee & Horng, 2020) and for rejecting interference (J. Wang et al., 2019). It is also used for identifying PD signals through clustering (B. Chen et al., 2017). An Adaptive Neuro-Fuzzy Inference System (ANFIS), which integrates neural networks with fuzzy logic, has been used for evaluating PD intensity by leveraging its ability to continuously optimize model parameters and reduce errors compared to traditional methods (J. Wang et al., 2019). The adaptability and robustness of ANFIS make it a strong candidate for PD analysis. Fuzzy logic methods are particularly valuable because they can be designed to provide interpretable, rule-based outputs, which aids in understanding the diagnostic decision-making process. Regarding performance, the Fuzzy Logic Clustering Decision Tree (FLCDT) was found to perform significantly better than traditional decision trees in terms of classification precision (Lee & Horng, 2020). An adaptive Fuzzy C-Means (FCM) method reported high accuracy in identifying PD signals (B. Chen et al., 2017). Furthermore, an improved ANFIS model showed a 2% reduction in model error compared to a traditional ANFIS when evaluating PD intensity (J. Wang et al., 2019).

2.2.7 Clustering Algorithms (K-Means, GMM, DBSCAN, etc.)

Clustering algorithms are unsupervised learning techniques that group objects such that those in the same cluster are more similar to each other than to those in different clusters. Common methods include K-Means, Gaussian Mixture Model (GMM), and density-based approaches like DBSCAN. Self-Organizing Maps (SOM) can also be viewed as a clustering technique. These algorithms have been applied to separate signals from multiple PD sources (Firuzi et al., 2017; Mishra et al., 2019) and perform data cleaning by separating signal from noise (Soh et al., 2022). They can also identify distinct PD patterns for localization (K. Zhou et al., 2021). Methods like K-Means, GMM, and Mean-shift have been combined to classify PD signals effectively (Carvalho et al., 2024). Self-Organizing Maps, when paired with statistical feature transformation, have been employed for PD classification (Bohari et al., 2021). A notable advancement is the development of PDAutoClust, a parameter-free clustering technique designed to reduce reliance on user-defined parameters, making the process more robust and accessible (Rahman et al., 2024). Clustering algorithms are valuable because they can discover underlying structures in PD data without requiring pre-labeled datasets. This is crucial for unsupervised monitoring where fault types may not be known in advance. GMM clustering helps discover distinct groups in a feature space for separating mixed PD signals (Firuzi et al., 2017), while density-based methods can effectively separate PD pulses from various types of noise and interference (K. Zhou et al., 2021). Performance-wise, a combination of K-Means, GMM, and Mean-shift clustering followed by an SVM classifier achieved 93% accuracy (Carvalho et al., 2024). The PDAutoClust method performed favorably against other density-based algorithms on PD datasets (Rahman et al., 2024). A SOM-based classifier demonstrated classification times under five seconds with near-zero error rates (Bohari et al., 2021), and an advanced density peak clustering algorithm reported an average localization error of just 0.21 m for two simultaneous PD sources (K. Zhou et al., 2021).

2.3 Feature Engineering in Partial Discharge Analysis

2.3.1 Time-Domain Features

Time-domain features are derived directly from the individual PD pulse waveforms to quantify their fundamental physical properties. These features characterize the shape, magnitude, and temporal occurrence of the pulse, which are intrinsically linked to the underlying discharge physics and provide crucial information for classification (Kumar et al., 2024; Lu et al., 2020). The analysis of these characteristics enables the differentiation of various PD phenomena based on how the discharge manifests as a transient electrical signal (Boczar et al., 2017).

The peak amplitude, signifying the maximum magnitude of the pulse, serves as a primary indicator of the discharge intensity and is directly related to the energy released during the event (K. Zhang et al., 2024). Different PD sources, such as corona, surface discharges, or internal voids, often generate pulses with distinct amplitude ranges. This makes amplitude a valuable feature for assessing a defect's severity and for distinguishing between source types, as the physical mechanism and geometry of the defect govern the amount of charge mobilized in the discharge (Boczar et al., 2017; Martínez-Tarifa et al., 2022).

The pulse rise time, defined as the interval for the signal to increase from 10% to 90% of its peak value, reflects the speed of the ionization process (Lu et al., 2020). PD pulses are characterized by very rapid rise times, often in the nanosecond or sub-nanosecond range (Darwish et al., 2019). This parameter is highly effective for discriminating between PD sources because phenomena such as streamer-like corona discharges exhibit different development speeds compared to Townsend-like discharges in internal voids (Xavier et al., 2021). Consequently, rise time is a critical feature, though it can be affected by signal degradation and dispersion during propagation from the source to the sensor (Lu et al., 2020).

Pulse duration, frequently measured as the full width at half maximum (FWHM), quantifies the time for which the discharge event persists and is related to the recombination time of charge carriers following the initial ionization (Martínez-Tarifa et al., 2022). This temporal characteristic is another key descriptor of the pulse shape that aids in distinguishing between defect types. Specific discharge mechanisms, influenced by factors like gas pressure and electrode geometry, produce characteristically longer or shorter pulses, making duration a useful parameter for classification (Boczar et al., 2017; Uwiringiyimana et al., 2022).

Finally, the pulse count per cycle tallies the total number of PD events detected within one period of the power frequency, directly quantifying the repetition rate and activity level of the PD source (Gillis et al., 2023). A high pulse count can indicate a more active or severe insulation defect, and monitoring this metric is essential for tracking the evolution and stability of a fault over time (Ma et al., 2021). The number of discharges per cycle, along with their phase distribution, is fundamental to forming Phase-Resolved Partial Discharge (PRPD) patterns, which are widely used for identifying specific defect types (Boczar et al., 2017; Kasinathan et al., 2017).

2.3.2 Statistical Features

Statistical features provide a quantitative description of the shape and distribution of discharge activity within the Phase-Resolved Partial Discharge (PRPD) pattern. These descriptors are derived from the distribution of pulse magnitudes or counts across the phase angles of the AC voltage cycle. For this work, the higher-order statistical moments of skewness and kurtosis were computed, as they are particularly effective for capturing the nuanced characteristics of PRPD patterns that differentiate various discharge phenomena (Boczar et al., 2017).

Skewness measures the degree of asymmetry in the distribution of PD pulses relative to the center of the phase axis. This feature is valuable because the physical mechanisms governing PD can differ significantly between the positive and negative half-cycles of the applied voltage, leading to characteristically asymmetric PRPD patterns (Boczar et al., 2017). For instance, corona from a sharp point electrode typically produces a much higher discharge activity during one polarity of the AC cycle, resulting in a highly skewed distribution (Yao et al., 2016). By quantifying this asymmetry, skewness provides a powerful metric for distinguishing between sources like corona, surface, and internal void discharges, each of which exhibits unique symmetry properties in its PRPD profile (Bohari et al., 2021).

Kurtosis quantifies the ”peakedness” or ”tailedness” of the distribution, indicating whether the PD pulses are concentrated within narrow phase windows or are more broadly dispersed across the cycle. A high kurtosis value signifies a distribution with sharp, narrow peaks and heavy tails, which is characteristic of discharge types that occur only at specific, repeatable moments near the voltage peaks (Boczar et al., 2017). Conversely, a lower kurtosis suggests a flatter, more uniform distribution of pulses. Since different insulation defects generate discharges with distinct phase concentrations, kurtosis serves as a key feature for identifying the nature of the PD source by characterizing the shape of its statistical distribution (Kunicki et al., 2018). Together, these statistical moments transform the qualitative visual information of a PRPD pattern into a compact and discriminative numerical feature vector suitable for automated classification (Bohari et al., 2021).

2.3.3 Frequency-Domain Features

Frequency-domain analysis is employed to investigate the spectral composition of partial discharge signals, which provides critical diagnostic information not apparent in the time domain. By applying a Fast Fourier Transform (FFT) to each captured waveform, the signal is converted from its temporal representation to its constituent frequencies (X. Li et al., 2018). The significance of this analysis lies in the fact that PD pulses are inherently fast, transient events with rapid rise times, meaning they are broadband phenomena containing significant energy across a wide spectrum that can extend into the High Frequency (HF), Very High Frequency (VHF), and Ultra-High Frequency (UHF) ranges (Darwish et al., 2019).

The frequency signature of a measured PD signal is shaped by two key factors: the physical nature of the discharge source and the propagation path the electromagnetic wave travels to the sensor (Azirani et al., 2021). The equipment itself, such as a power transformer or a Gas-Insulated Switchgear (GIS) enclosure, acts as a complex waveguide, causing frequency-dependent attenuation and reflection that alters the signal's spectral characteristics (Promwong & Tiengthong, 2020). Consequently, the energy distribution across different frequency bands becomes a unique fingerprint for a specific defect type within a given apparatus.

Different types of insulation defects are known to manifest distinctively in the frequency domain. The UHF band (300 MHz–3 GHz) is particularly valuable for PD detection in transformers and GIS, as signals in this range can propagate efficiently within the metallic enclosures while being less susceptible to external corona and broadcast interference (Salah et al., 2022). For instance, the spectral content of UHF signals can be analyzed to diagnose and distinguish between specific fault types in GIS, such as those caused by free-moving particles or fixed protrusions on conductors (Ren et al., 2021). Selecting an optimal frequency band for analysis is therefore crucial for maximizing the signal-to-noise ratio and detection sensitivity (Azirani et al., 2021). For this study, the spectral power was computed within the distinct HF, VHF, and UHF bands to create a feature that captures this discriminative energy distribution, providing a robust basis for classification (Maneerot et al., 2019).

2.3.4 Time-Frequency Features

Partial discharge signals are fundamentally transient and non-stationary phenomena, meaning their frequency content evolves rapidly over their short duration. This characteristic makes time-frequency analysis methods, which can capture both temporal and spectral information simultaneously, particularly well-suited for their characterization (X. Li et al., 2018). Among these methods, the Discrete Wavelet Transform (DWT) is exceptionally effective and widely employed in PD analysis for its dual capability of signal denoising and feature extraction (Lu et al., 2020). The DWT decomposes a signal into various frequency sub-bands at different resolutions, allowing for the effective isolation of PD-related components from background noise and interference (Sahnoune et al., 2024). The efficacy of the DWT is highly dependent on the choice of the mother wavelet, as its shape should ideally correlate with the morphology of the PD pulses being analyzed to achieve optimal decomposition (Sahnoune et al., 2024). For this study, the Daubechies 4 (db4) wavelet was selected. The Daubechies family of wavelets is frequently chosen for its properties of compact support and orthogonality, and the db4 wavelet, in particular, has been demonstrated in the literature to be effective for the denoising and analysis of PD signals, thereby representing an established and validated choice (G. Wang et al., 2020). The signal was decomposed into four levels, which separates the signal into a series of detail coefficients representing successively narrower high-frequency bands, and a final approximation coefficient representing the low-frequency content. For feature extraction, the energy of the detail coefficients at each of the four decomposition levels was calculated. The resulting four-dimensional vector of wavelet energies serves as a powerful feature set. This approach is justified because the energy of the coefficients at a specific level quantifies the intensity of the signal within that particular frequency band (Yang et al., 2024). Since different types of PD sources (e.g., surface, corona, void) generate pulses with distinct spectral signatures, their energy is distributed differently across these frequency bands (Gao et al., 2023). Therefore, the vector of wavelet energies creates a compact and highly discriminative signature that effectively characterizes the unique time-frequency energy distribution of a PD event, making it an excellent input for machine learning classifiers (Yang et al., 2024).

2.4 Dataset Diversity and Model Generalizability

The generalizability of machine learning (ML) models for Partial Discharge (PD) detection is fundamentally challenged by the immense diversity of PD data (Lu et al., 2020). Signal characteristics are highly sensitive to the defect type (Lee & Horng, 2020), insulation material (Zang et al., 2022), specific HV apparatus (Bin et al., 2020; Meitei et al., 2020), and operational environment. Consequently, a model trained on a specific dataset, particularly laboratory-generated data, often fails to generalize to different real-world scenarios (Raymond et al., 2022; Sun et al., 2019).

To mitigate this, researchers employ strategies like domain adaptation (Z. Li et al., 2024; Niu & Zhu, 2025), federated learning (Tai et al., 2024; Tuyet-Doan et al., 2024), and data augmentation to improve model robustness (Rauscher et al., 2024; Raymond et al., 2022). Despite these efforts, a critical gap persists: many studies still rely on limited, homogeneous datasets from controlled lab settings (Q. Zhang et al., 2016). The scarcity of real-world data and the absence of standardized public benchmarks severely hinder the development of truly robust and widely applicable ML solutions (J. Kim & Park, 2020; Lu et al., 2020).

2.5 Comparative Studies and Benchmarking in ML for Partial Discharge Detection

Comparative studies and benchmarking are essential for identifying the most effective algorithms and methodologies in ML-based PD detection. Several studies have undertaken such comparisons, revealing performance differences among models. For example, (Bin et al., 2020) found that a PSO-ELM outperformed SVM and k-NN for GIS defect classification. (Lee & Horng, 2020) showed their Fuzzy Logic Clustering Decision Tree surpassed standard decision trees for transformer diagnostics. Likewise, (Ma et al., 2021) demonstrated that SVM was superior to k-NN, Naive Bayes, and Decision Trees for classifying PD severity.

However, existing comparative analyses often suffer from inconsistencies that make direct, field-wide comparisons challenging (Lu et al., 2020). A primary issue is the use of different, often private, datasets, which limits the generalizability of findings (J. Kim & Kim, 2021). Since PD signal characteristics vary significantly with the defect type, insulation, and equipment, results from one dataset are not directly applicable to another (Ranjan et al., 2024).

A key limitation is that comparisons often focus on algorithmic performance without deeply exploring how feature engineering strategies interact with the choice of model (Govindarajan et al., 2021). The optimal ML algorithm for a given task might change depending on the features extracted from the raw signal. There is a notable lack of standardized benchmarking protocols that test across various feature-algorithm pairings and generalization scenarios (Lu et al., 2020). This need for standardization was highlighted in the development of a qualification procedure for PD analysers (Vera et al., 2023). This absence means that models risk being overfitted to specific datasets (Raymond et al., 2022), the evaluation of algorithm-feature combinations is limited (Kumar et al., 2024), and model generalizability across diverse real-world conditions is often insufficiently assessed (Lu et al., 2020). Addressing these issues is essential for developing truly robust and reliable ML-based PD detection systems.

2.6 Research Gaps

Despite the promise of machine learning (ML) for Partial Discharge (PD) detection, a review of the literature reveals critical gaps that hinder the development of robust and generalizable diagnostic tools. A primary gap is the limited exploration of feature-algorithm interactions; most studies evaluate ML models using a single, predefined set of features (Kumar et al., 2024; Lu et al., 2020), neglecting to systematically investigate how different feature combinations impact the performance of various algorithms (Govindarajan et al., 2021; Nimmo et al., 2017). Furthermore, many models are trained and tested on limited, homogeneous datasets from a single equipment type or condition (J. Kim & Kim, 2021; Woon et al., 2016). This practice raises significant questions about their ability to generalize to unseen PD sources and diverse operational environments, a key barrier to practical deployment (Lu et al., 2020).

This research addresses these gaps by systematically comparing traditional ML models across diverse PD datasets. We will evaluate how various feature combinations affect each algorithm's performance to identify optimal pairings, a need highlighted by the lack of such analysis in current literature (Govindarajan et al., 2021). Furthermore, we will explicitly test model generalizability using a combined dataset reflecting real-world conditions from various equipment like transformers, GIS, and cables (Bin et al., 2020; Meitei et al., 2020), aiming to develop more robust and reliable diagnostic tools.

CHAPTER 3

METHODOLOGY

This chapter details the systematic approach undertaken to investigate Partial Discharge phenomena. It encompasses the overall research design, the acquisition and nature of datasets, the methodologies for feature extraction from partial discharge signals, the classification techniques employed for identifying partial discharge sources or types, and the metrics used for evaluating the performance of these techniques. A rigorous methodology is paramount for ensuring the reliability and validity of research findings in the complex domain of partial discharge analysis.

3.1 Overview of the Proposed Methodology

The methodology for partial discharge research generally follows a structured pipeline. This typically begins with the acquisition or sourcing of partial discharge signal data, often from experimental setups or simulations mimicking real-world conditions such as defects in power cables, transformers, or Gas-Insulated Switchgear (Borghei & Ghassemi, 2022). These signals are frequently contaminated with noise, necessitating robust signal processing and denoising techniques (Friebe & Jenau, 2024; Ghosh et al., 2020; Sun et al., 2019). Subsequently, relevant features are extracted from the cleaned signals in various domains (time, frequency, time-frequency) to create a discriminative representation of the discharge event (Bag et al., 2019; Kumar et al., 2024). These features then serve as input to machine learning classifiers or other analytical models to identify partial discharge types, locate their sources, or assess insulation degradation (Aldosari et al., 2023; Anjum et al., 2017; Rauscher et al., 2024). Finally, the performance of these models is rigorously evaluated using appropriate metrics to ascertain their accuracy and reliability in practical diagnostic scenarios (Srivastava et al., 2023; Yu et al., 2024).

Building upon this established framework, the present study implements a specific methodology designed for the comparative analysis of traditional machine learning algorithms. This research leverages a unique, long-term dataset acquired through the contactless antenna-based monitoring of partial discharges on covered overhead power lines (Klein et al., 2023, 2024). A core component of our approach is the extraction of a comprehensive suite of features from multiple domains, a strategy widely advocated for enhancing diagnostic accuracy (Lu et al., 2020). These features are then systematically used to train and test a selection of established classifiers in a comparative framework (Amaya-Sanchez et al., 2024). The ultimate goal is to identify optimal feature-algorithm pairings for this specific application and to critically assess the generalizability of the resulting models, thereby providing insights into robust diagnostic solutions for real-world field conditions (Seitz et al., 2024). The subsequent sections of this chapter will detail each of these stages: the dataset and preprocessing steps (Section 3.2), the feature engineering process (Section 3.3), the selection and implementation of machine learning classifiers (Section 3.4), and the protocol for performance evaluation (Section 3.5).

3.2 Datasets

The research on Partial Discharge (PD) analysis and detection is fundamentally driven by data, and this work specifically utilizes a unique dataset from the long-term, on-line monitoring of medium-voltage (MV) overhead power lines. This approach aligns with a growing trend in the field that moves beyond purely laboratory-based experiments to include data captured from in-situ operational equipment, which presents both distinct challenges and opportunities for developing robust diagnostic systems (S. Kim et al., 2020; Sikorski & Gielniak, 2025).

The primary dataset for this research was obtained using a contactless antenna method designed specifically for detecting partial discharges in XLPE-covered conductors. The use of antennas for PD detection is a well-established non-invasive technique that captures the electromagnetic waves radiated by discharge events, allowing for remote monitoring without physical contact with the high-voltage components (Azam et al., 2024; Cruz et al., 2023). This radiometric approach is particularly advantageous for monitoring widespread assets like overhead lines, as it circumvents the need for direct sensor installation on every component. Our focus on overhead lines with covered conductors is supported by a significant body of research that has specifically investigated fault detection and PD phenomena in these systems, acknowledging their importance for overall grid reliability and the unique challenges they present (K. Chen et al., 2021; C. Zhang et al., 2024).

The dataset is characterized by its extensive temporal scope and geographical distribution. It comprises nearly three years of continuous data, collected at one-hour intervals from nine different measuring stations situated in Czechia and Slovakia. Each sample within this dataset is a time-series signal representing a 20-millisecond snapshot of the electromagnetic activity captured by the antenna. This long-term, continuous data acquisition from multiple sites provides a rich source of information, capturing a wide variety of PD signals under diverse environmental and operational conditions. The ability to track pattern evolution over extended periods is crucial for prognostic health management and anomaly detection (Florkowski, 2024; Orellana et al., 2023).

A key strength of our dataset is the availability of reliable ground truth information, which is essential for the development and validation of diagnostic models. The contactless antenna systems were deployed at the same monitoring stations where a traditional galvanic contact method, used by power distribution companies, is also in operation. This parallel measurement provides a direct, industry-accepted reference for confirming PD activity. Furthermore, the dataset has been enhanced with manually curated labels provided by human experts. This expert validation is invaluable for training and assessing the performance of supervised machine learning algorithms, ensuring that the models learn to identify genuine PD events accurately. The availability of such high-quality, labeled data enables the application of various supervised machine learning models, such as Support Vector Machines (SVM) and ensemble methods like Random

Forests, to achieve high-accuracy fault detection and classification (Govindarajan et al., 2021; Serttaş & Hocaoğlu, 2020). The large volume of data also necessitates research into efficient data handling, such as novel lossy compression methods and deployment on edge computing devices for real-time, on-site analysis (Klein et al., 2024).

3.3 Feature extraction

To facilitate the classification of partial discharge (PD) signals, a critical stage in the diagnostic pipeline is the transformation of raw, preprocessed time-series data into a concise and discriminative set of numerical features (Kumar et al., 2024). For this study, a multi-domain feature extraction strategy was employed, encompassing descriptors from the time, statistical, frequency, and time-frequency domains. This comprehensive approach aims to construct a robust representation of PD events by capturing the distinct signal characteristics essential for accurate diagnosis by machine learning classifiers (Lu et al., 2020; Mas'ud et al., 2016). The subsequent sections provide a detailed account of the features calculated for this purpose. The feature that will be used in this are Time-domain feature, Statistical feature, Frequency-Domain Features, Time-frequency Features

3.4 Classification and Cross-Validation

This section outlines the classification algorithms employed in this study and the strategies used to validate their performance. Emphasis is placed on selecting methods that align with the complex and diverse nature of partial discharge data, followed by a cross-validation framework to ensure rigorous model assessment.

3.4.1 Algorithm Selection and Justification

We checked for problems in electrical insulation by studying the signals from tiny sparks, known as partial discharges. To do this, we used five different computer models to analyze the spark signals and identify the type of insulation problem. We chose a variety of models to see which method would work best.

A Support Vector Machine (SVM) was used because this model is very good at finding the best possible line to divide one type of problem from another, even when the data is complicated.

A Random Forest model was also used. This method builds many simple decision flowcharts and then gets them to vote on the answer. This ”teamwork” approach makes it more accurate and reliable.

The K-Nearest Neighbors (KNN) model works by finding the most similar examples from the past. It assumes a new problem is the same type as its closest ”neighbors.”

We used a single Decision Tree because it is very easy to understand. It creates a straightforward flowchart with a series of basic ”if-this-then-that” questions to figure out the problem.

Finally, an Artificial Neural Network (ANN), a model inspired by the human brain, was included. We used it because it is excellent at finding very complex and hidden patterns in the data that other models might miss.

3.4.2 Model Validation Strategy

To ensure a robust and unbiased evaluation of model performance, a k-fold cross-validation strategy was employed. This validation technique is essential for mitigating the risk of overfitting, particularly with limited experimental datasets, and for providing a reliable estimate of each model’s generalization capability on unseen data. It is considered a standard procedure for assessing machine learning models in PD monitoring systems (Srivastava et al., 2023) and a critical component in the qualification process for PD analysers (Vera et al., 2023). In this study, 10-fold cross-validation was utilized. The dataset was partitioned into ten mutually exclusive subsets, or ‘folds’, of approximately equal size. The classification process was then iterated ten times; in each iteration, one distinct fold was reserved as the test set for validation, while the remaining nine folds were used for training the model. The final performance metrics for each algorithm were calculated by averaging the results obtained across all ten folds. This approach ensures that the reported performance is a stable and comprehensive reflection of the model's diagnostic capabilities.

3.5 Evaluation Metrics

A comprehensive and rigorous evaluation of classifier performance is essential for developing reliable diagnostic systems. The assessment of the proposed partial discharge (PD) classification models will be conducted using a suite of standard metrics, each chosen to provide a distinct perspective on the model's effectiveness. This multifaceted approach ensures that the evaluation captures not only the overall correctness but also the specific strengths and weaknesses related to identifying individual fault types, as well as the practical feasibility for deployment in monitoring systems.

Confusion Matrix

The confusion matrix is a fundamental tool that provides a detailed breakdown of a classifier's performance on a class-by-class basis. It is structured as a table that explicitly visualizes the counts of correct and incorrect predictions, tabulating the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN) for each diagnostic category. This matrix serves as the bedrock for calculating most other performance metrics and offers a direct, qualitative analysis of misclassifications (Sikorski & Gielniak, 2025). In the complex context of PD diagnosis, the confusion matrix is invaluable because it reveals not just that errors occurred, but specifically which defect types are being confused with one another. This granular insight is critical for understanding the diagnostic limitations of a model and for guiding future improvements, such as refining feature extraction to better distinguish between similar fault signatures.

Accuracy

Accuracy represents the proportion of all correctly classified instances relative to the total number of instances evaluated, calculated as (TP + TN) / (TP + TN + FP + FN). It serves as a primary, high-level indicator of a model's overall performance and is a fundamental benchmark metric used extensively throughout PD classification research (Aldosari et al., 2023). While it provides an intuitive and useful general summary of correctness, its diagnostic utility can be limited, particularly when dealing with the imbalanced class distributions that are characteristic of fault diagnosis data. In such scenarios, a model can achieve a high accuracy score by simply predicting the majority class (e.g., 'no fault'), while failing to identify rare but critical fault conditions. Therefore, it is always interpreted in conjunction with more nuanced, class-specific metrics.

Precision

Calculated as TP / (TP + FP), precision quantifies the exactness of the classifier by measuring the proportion of instances classified as positive that are genuinely positive. For PD diagnostics, high precision is essential for ensuring the reliability and trustworthiness of a fault warning. This metric directly correlates to minimizing the false alarm rate, which is a critical operational objective. A high-precision diagnostic system prevents unnecessary and costly maintenance interventions, avoids unwarranted equipment downtime, and builds operator confidence in the automated monitoring system's outputs, making it a key indicator of its practical value (Y. Li et al., 2025).

Recall (Sensitivity)

Calculated as TP / (TP + FN), recall, also known as sensitivity or the true positive rate, measures the classifier's ability to correctly identify all actual instances of a specific fault class. Within the domain of high-voltage asset management, achieving a high recall is paramount, as it reflects the model's completeness in fault detection. Failing to detect a genuine PD source—a false negative—can have severe consequences, potentially leading to progressive insulation degradation, unexpected outages, and ultimately, catastrophic equipment failure (Adhikari et al., 2024). This metric is therefore a critical indicator of the system's effectiveness in proactive risk mitigation and ensuring the safety and reliability of the power network.

F1-Score

The F1-Score is computed as the harmonic mean of precision and recall, using the formula 2 * (Precision * Recall) / (Precision + Recall). It provides a single, balanced measure of a model's performance by considering both false positives and false negatives. It is particularly valuable for evaluating classifiers on imbalanced datasets, which is a common scenario in fault diagnosis where data from certain critical fault types may be scarce compared to normal operating data (Zemouri & Levesque, 2023). By balancing the inherent trade-off between precision and recall, the F1-Score offers a more robust and comprehensive assessment of a model's ability to correctly classify all categories, preventing a model's high performance on the majority class from masking its poor performance on infrequent but critical fault classes (Y. Li et al., 2025).

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